

Short-term durability test for GFRP rods under various environmental conditions

Hyeong-Yeol Kim ^{a,*}, Young-Hwan Park ^a, Young-Jun You ^a, Chang-Kwon Moon ^b

^a Structure Research Department, Korea Institute of Construction Technology, 2311 Daewha-Dong Ilsan-Gu Goyang, Gyeonggi-Do 411-712, Republic of Korea

^b Department of Materials Science and Engineering, Pukyong National University, 100 Yongdang-Dong, Busan 608-739, Republic of Korea

Available online 20 March 2007

Abstract

This paper presents the results of short-term durability test for GFRP rods exposed to various environmental conditions. The main objective of this study is to investigate the degradation of GFRP composite using an accelerated aging method and its influence on the tensile properties. Two types of E-glass/vinylester rods were exposed to moisture, chloride, alkali, and freeze-thaw cycling conditions for up to 132 days. In order to examine the degradation of the GFRP composite in micro level, strand-type E-glass/vinylester rod specimens were also fabricated and tested. The test results indicated that the tensile properties of the GFRP rods were significantly reduced after conditioning, due to the degradation of GFRP.

© 2007 Elsevier Ltd. All rights reserved.

Keywords: Durability; Glass fiber; Thermosetting resin; Tensile test; Accelerated aging

1. Introduction

The corrosion of steel reinforcement and the cracking of concrete are the main causes of the deterioration and structural deficiency of reinforced concrete structures. To overcome the corrosion problem of steel reinforcement in concrete, the steel reinforcing bar (rebar) should be coated or replaced with non-corrosive materials.

In recent years, fiber reinforced polymers (FRP) have been widely used for civil infrastructures throughout the world, especially in the fields where light-weight, high strength, and high durability are required [1,2]. High resistance to the corrosion of FRP may be an ideal characteristic for the materials used in civil structures under the corrosive environments. In the late 1980s, an FRP rebar was introduced as a new type of internal reinforcement for concrete bridge decks. Currently, several different types of FRP bars are commercially available as the substitutes

for steel reinforcement in concrete structures. A large number of bridge decks and other structural elements reinforced with FRP bars are already in service and several projects are currently underway [2–5].

Among the fibrous materials, glass fibers are most widely used for civil structures due to their low cost. However, glass FRP (GFRP) composites are prone to degrade when they are exposed to certain environmental conditions. In particular, it is known that glass fibers are degraded by moisture and alkali due to etching and leaching actions [6–8]. The degradation of matrices occurs due to hydrolysis, plasticization, and swelling in the presence of water. This also weakens the interface between the glass fibers and the matrix, and decreases the tensile properties of GFRP composites.

If GFRP composites are used for the civil structures, the influence of water on the durability of the GFRP composites is a critical issue since the civil structures are usually exposed to moisture during their service life. The influence of alkali on the durability of GFRP composites is also important because the concrete is a high alkaline material.

* Corresponding author. Tel.: +82 31 910 0582; fax: +82 31 910 0121.
E-mail address: hykim1@kict.re.kr (H.-Y. Kim).

However, few long-term performance data are available for GFRP composites to date [9–11]. Therefore, the durability performance of GFRP composites has been an important issue in recent years.

During the last two decades, the durability of GFRP composites related to civil infrastructure applications has been extensively investigated and the highlights of results of these studies are well summarized in [6,8,9,12]. The environmental degradation of GFRP composites due to various environments such as moisture, alkali, chloride, temperature, and wetting and drying cycles and its influence on the tensile and bond properties of GFRP composites are the main interests.

A number of researches [8,12–14] have studied the durability performance of commercially available GFRP (E-glass/vinylester) bars exposed to various environmental conditions. Generally, an accelerated aging method with elevated temperature was used to accelerate the degradation process. The influence of high temperature with and without long-term exposure of alkaline environments on the durability of GFRP rebars was studied by Abbasi and Hogg [15]. The combined effects of sustained stresses and moisture on the durability of GFRP composite were also studied by Abdel-Magid et al. [16]. They have found that that moisture and alkali significantly affect not only the degradation of glass fiber and matrix but the interface between the fiber and the matrix.

To investigate the degradation of the material properties of GFRP bars with time, several creep rupture tests with and without environmental conditions have been conducted [17–19]. Several experimental studies [6,14,20–23] have been reported in the literatures for the bond degradation of GFRP bars exposed to various environmental conditions. Since the concrete flexural members reinforced with GFRP rebars undergo large deflection as compared to those reinforced with steel rebars due to the low elastic modulus of GFRP, a number of researches [24–29] have studied the flexural behavior of concrete beams reinforced with GFRP rebars.

This paper deals with short-term durability test for bare GFRP (E-glass/vinylester) rods exposed to various environmental conditions. The main objective of this study is to investigate the degradation of the GFRP composite using an accelerated aging method and its influence on the tensile properties. The environmental degradation at the interface between the GFRP rod and concrete and the time-dependent characteristics due to load are beyond of the scope of this study.

The durability test program presented in this paper consists of two parts by size of specimen, a full-size specimen and a strand-type specimen. In the test program with full-size specimens, two types of commercially available GFRP (E-glass/vinylester) rebar products of 12.7 mm in diameter were exposed to four different environmental conditions for up to 132 days. On the other hand, to examine the degradation of GFRP rods in micro level, strand-type E-glass/vinylester specimens of 0.7 mm in diameter were

pultruded and were exposed to five different environmental conditions for up to 150 days.

The influence of environmental factors on the degradation of GFRP was evaluated by comparing the tensile properties of the unconditioned and conditioned specimens. Scanning electronic microscope (SEM) was also used to observe the physical condition and damage in the GFRP composites after the tensile test.

2. Test program for full-size specimens

To quantitatively evaluate the durability characteristic of GFRP rods, two types of pultruded GFRP rods named herein as G1 (E-glass/vinylester) and G2 (E-glass/modified vinylester) were selected and tested. The mechanical properties and fiber volume fraction of as-received GFRP rods were tested and the results are summarized in Table 1. Tensile and interfacial shear tests were carried out in accordance with ASTM D 3916 [30] and ASTM D 4475 [31] and their typical test set-ups are shown in Figs. 1 and 2, respectively. The elastic modulus of the specimen was estimated by using the stress–strain relationship obtained in the tensile test.

Table 1
Mechanical properties of GFRP rods

Rod type	Tensile strength (MPa)	Elastic modulus (GPa)	Shear strength (MPa)	Fiber volume fraction by weight (%)
G1	692.8	38.5	36.9	73.3
G2	660.6	42.7	42.3	69.1

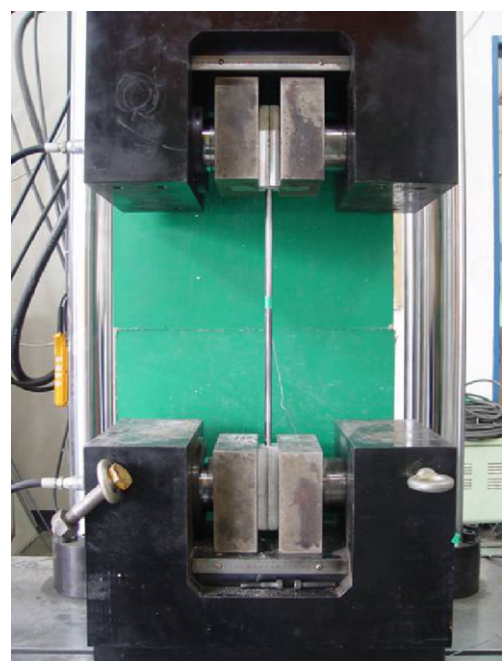


Fig. 1. Set-up for tensile test.

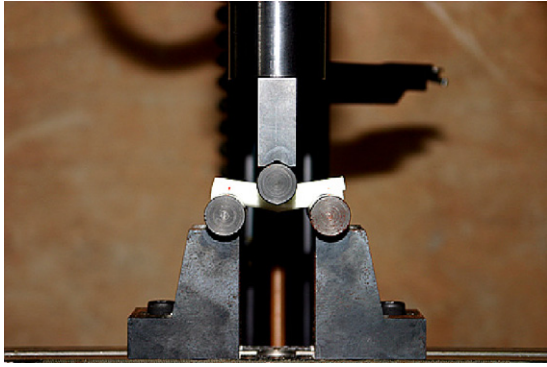


Fig. 2. Set-up for interfacial shear test.

Table 2
Aging methods for full-size specimens

Condition	Solution	Temperature (°C)	Duration (days)
Moisture	Tap water	25, 40, 80	30, 60, 90, 132
Chloride	3% NaCl solution	25, 40, 80	30, 60, 90, 132
Alkali	Alkaline solution (pH13)	25, 40, 80	30, 60, 75
Freeze-thaw	Alkaline solution (pH13)	−25 to 30	110

2.1. Aging and testing methods

The aging methods used for the durability test are summarized in Table 2. In practice, the FRP rebar embedded in concrete structures can be exposed to moisture, chloride, alkali, and freeze-thaw cycling conditions. Tap water was used to simulate high humidity and moisture. Three-percent NaCl solution was used to simulate the chloride environmental condition caused by sea water. Solution which contains 1.4% KOH, 1.0% NaOH, and 0.16% Ca(OH)₂ by weight was used for the alkaline condition to simulate concrete pore solution with high pH level (pH 13). In addition to the room temperature (25 °C), elevated temperatures of 40 °C and 80 °C were employed to accelerate the degradation of the GFRP rods. For a freeze-thaw condition, temperature cycling from −25 °C to 30 °C was

repeated a day. The accelerated aging method employed in this test program is similar to the test protocols used in [8,14].

Three 900-mm-long specimens were placed in a galvanized steel tube of 50 mm in diameter that is filled with each simulated solution. The steel tubes were sealed and then immersed into 25 °C, 40 °C, and 80 °C water baths. A total of 264 GFRP rod specimens were exposed to four different environmental conditions.

In this test program, the maximum duration for all conditioning methods was originally appointed as 130 days. However, the conditioning in the alkaline solution was stopped on 75th day since the solution leakage from the sealed tubes was detected during the conditioning. The number of specimens used in the tensile and interfacial shear tests was three and eight, respectively.

2.2. Test results and discussion

Fig. 3(a) and (b) shows Specimens G1 and G2 that were immersed in the alkaline solution for 60 days, respectively. Fig. 4(a) shows a typical scanning electron microscope (SEM) image of an unconditioned Specimen G2, whereas Fig. 4(b) shows that of a conditioned Specimen G2 in the alkaline solution at 40 °C for 60 days. As shown in Fig. 4(b), all the edges of the fibers within the conditioned specimen were damaged during the polishing process for SEM. It is believed that this damage was mainly caused by the alkali and moisture attack, which also led to a weak interface between the fiber and the matrix. However, the degradation mechanism of GFRP rods cannot be directly identified from the SEM images.

Fig. 5 shows the tensile strength retention of the specimens as the function of immersion time and temperature. In the figure, the tensile strength of the conditioned specimen is divided by that of the unconditioned specimen and the corresponding value is denoted as the retention ratio in percent. Although there were some differences in the degradation pattern of the GFRP rods by influencing factor and specimen type, the tensile strength was reduced in all cases as the immersion time and temperature increased. The degradation patterns of the GFRP rods in

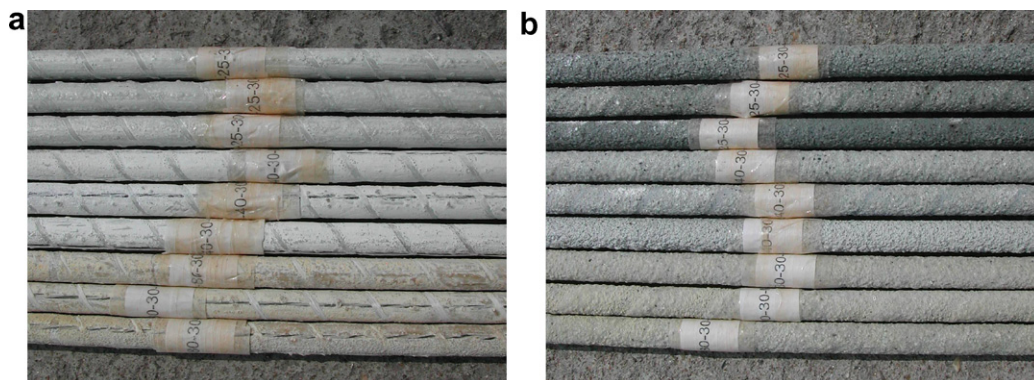


Fig. 3. Photos of conditioned specimens: (a) G1 and (b) G2.

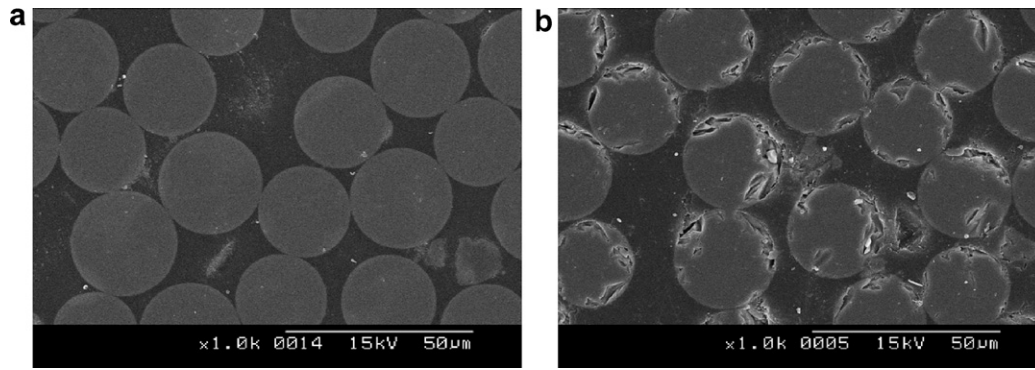


Fig. 4. SEM images of Specimen G2: (a) unconditioned and (b) immersed in alkaline solution at 40 °C for 60 days.

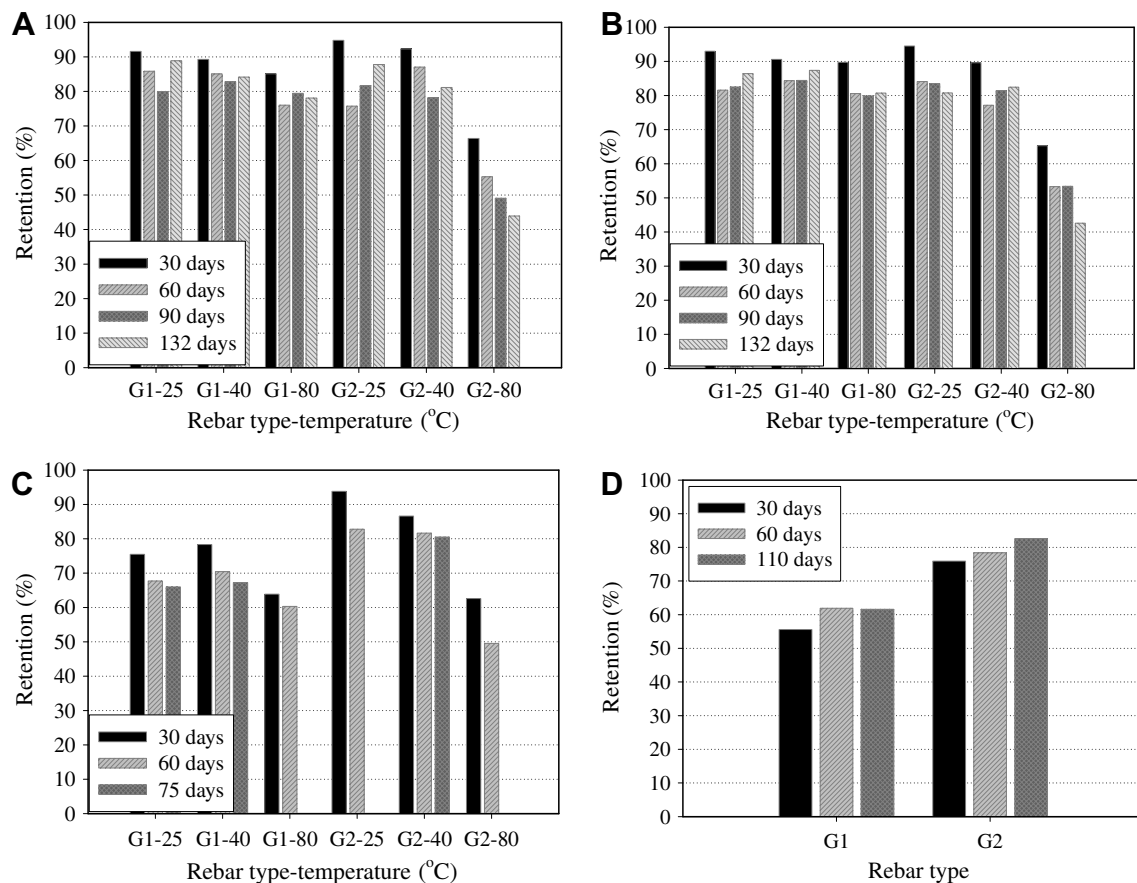


Fig. 5. Retention of tensile strength of full-size specimens conditioned in (a) water; (b) NaCl solution; (c) alkaline solution; and (d) freeze-thaw cycling.

the moisture and chloride conditions were found to be similar in nature for both G1 and G2. On the other hand, a significant strength loss was identified, especially for Specimen G1 immersed in the alkaline solution, regardless of temperature used.

For Specimen G2 conditioned at 80 °C, the tensile strength was generally reduced more than 50% regardless of the solution used. The vinylester resin used for G2 may have lower temperature resistance than that used for G1. Elevated temperature of 80 °C may be too high for Specimen G2. The strength reduction of the specimens con-

ditioned in the freeze-thaw cycling condition was similar to that in the alkaline condition. This reveals that the strengths in both cases were reduced mainly due to the alkali attack.

Table 3 provides the retention of tensile strength of the specimens immersed in tap water, along with the results of tests by other researchers [8,11,14]. Although material compositions of the specimens and environmental conditions used by others are somewhat different, similar degradation behavior was observed. Chu et al. [11] and DeJke [8] reported that the reduction of the tensile strength of GFRP

Table 3
Results of tensile test for specimens immersed in tap water

Temperature (°C)	Duration (days)	Retention of tensile strength (%)				
		G1	G2	Chu et al. [11]	Chen et al. [14]	Dejke [8]
23	70			92.4		
25	30	91.6	94.8			
	60	85.8	75.7			
	90	80.0	81.7			
	132	88.9	87.8			
40	25					91.7
	30	89.3	92.3			
	60	85.1	87.1			
	70			70.9	97.0	
	88					81.9
	90	82.8	78.2			
	132	84.1	81.2			
60	70			58.8	71.0	
80	21					67.2
	30	85.1	66.4			
	60	76.0	55.3			
	70			48.0		
	88					50.1
	90	79.5	49.1			
	132	78.1	43.9			

composites at 80 °C was 52% for 70 days and 49.9% for 88 days, respectively. These are compared to 24.0% and 46.7% for Specimens G1 and G2 for 60 days at 80 °C, respectively.

In Table 4, the retention of tensile strength of the specimens immersed in NaCl solution was compared with the test data by Chen et al. [14]. Chen et al. reported that the strength retention of GFRP bars immersed in NaCl solution for 70 days at 40 °C and 60 °C was 98% and 74%, respectively. On the other hand, the strength retention of Specimens G1 and G2 for 90 days at 40 °C was 84.4% and 81.5%, respectively.

Table 4
Results of tensile test for specimens immersed in NaCl solution

Temperature (°C)	Duration (days)	Retention of tensile strength (%)		
		G1	G2	Chen et al. [14]
25	30	93.0	94.5	
	60	81.6	84.1	
	90	82.5	83.5	
	132	86.4	80.8	
40	30	90.6	89.6	
	60	84.3	77.2	
	70			98.0
	90	84.4	81.5	
	132	87.4	82.5	
60	70			74.0
80	30	89.7	65.3	
	60	80.6	53.3	
	90	80.0	53.4	
	132	80.7	42.6	

Table 5 provides the retention of tensile strength of the specimens immersed in alkaline solution, along with the results of tests by other researchers [8,11,14]. The reduction of tensile strength for Specimens G1 and G2 for 60 days at 80 °C was 39.7% and 50.5%, respectively. These are compared to 49.1% for 88 days [8] and 53.7% for 70 days [11].

Fig. 6 shows the retention of elastic modulus of the specimens as the function of immersion time and temperature. Overall, the environmental conditions do not have significant influence on the elastic modulus of GFRP rods as on tensile strength. It is possibly said that the degradation of GFRP has less influence on the elongation of glass fiber. The elastic modulus was reduced by up to 25% for both G1 and G2. Some specimens showed increasing tensile property under the longer immersion time, as also reported in [8,13]. This may be because the test data are scattered but clear explanation cannot be made in this study.

The degradation of interface between the fiber and the matrix in GFRP can be evaluated by an interfacial shear test. The selected specimens were tested for the interfacial shear strength and the results are summarized in Table 6, along with the test data by others [8,13,14]. Except for Specimen G2 conditioned in the freeze-thaw cycling condition, there was obvious reduction in the interfacial shear strength in all influencing factors. The strength retention of Specimens G1 and G2 immersed in alkaline solution for 60 days at 40 °C was 83.2% and 76.1%, respectively. Dejke [8] and Chen et al. [14] reported that retention of the interfacial strength was 91.7% for 59 days at 60 °C and 92.0% for 45 days at 60 °C. On the other hand, the strength retention of 70% was observed by Micelli et al. [13] for 42 days at 60 °C.

Table 5
Results of tensile test for specimens immersed in alkaline solution

Temperature (°C)	Duration (days)	Retention of tensile strength (%)				
		G1	G2	Chu et al. [11]	Chen et al. [14]	Dejke [8]
23	70			86.3		
25	30	75.5	93.8			
	60	67.7	82.8			
	75	66.0				
40	21					73.7
	30	78.3	86.6			
	60	70.4	81.6			
	70			68.8	89.0	
	75	67.3	80.5			
	88					74.5
60	70			55.9	64.0	
	75					66.3
80	21					63.2
	30	63.9	62.6			
	60	60.3	49.5			
	70			46.3		
	88					50.9
	90					
	132					

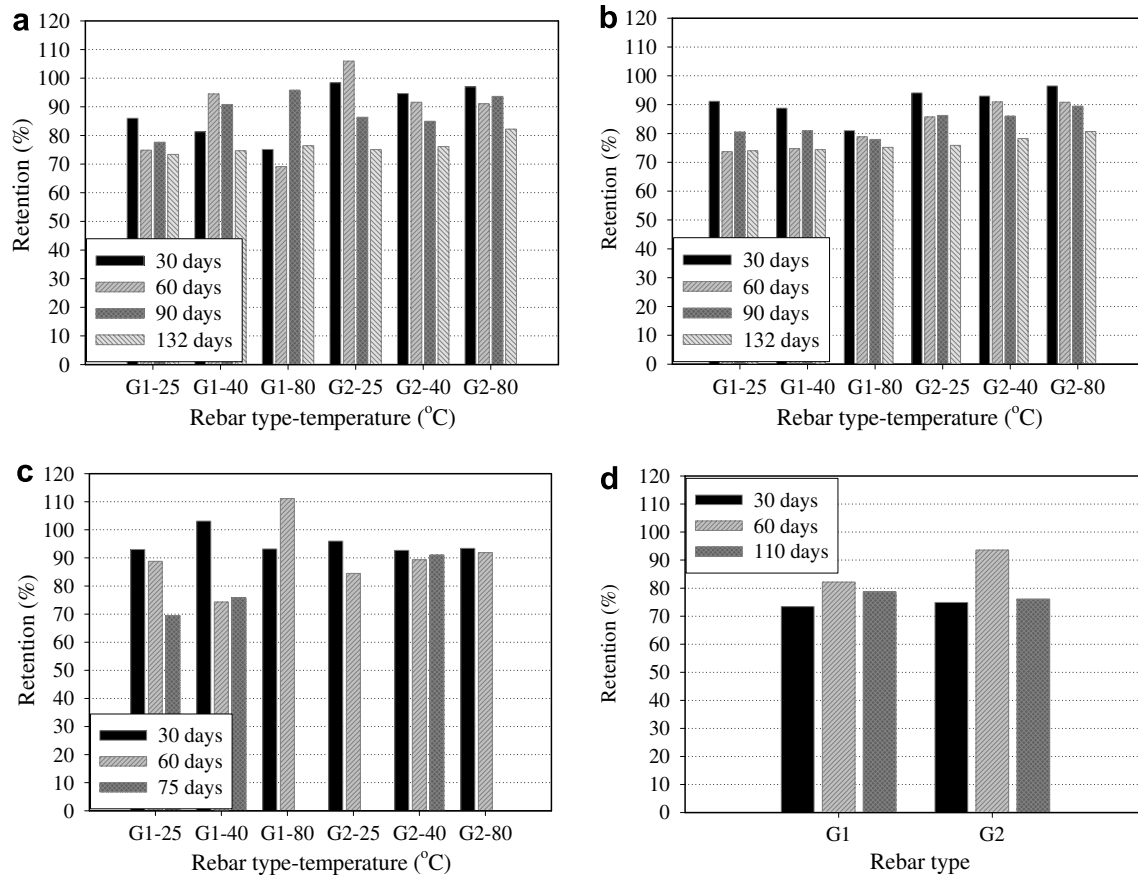


Fig. 6. Retention of elastic modulus of full-size specimens conditioned in (a) water; (b) NaCl solution; (c) alkaline solution; and (d) freeze-thaw cycling.

Table 6
Interfacial shear strength of specimens (retention in %)

Condition	Temperature (°C)	Duration (days)	Interfacial shear strength				
			G1	G2	Chen et al. [14]	Dejke [8]	Micelli et al. [13]
Moisture	60	45			96.0		
		59				92.2	
	80	90	89.4	76.6			
Alkali	40	60	83.2	76.1			
	60	42					70.0
		45			92.0		
		59				91.7	
Chloride	80	90	87.5	79.9			
Freeze-thaw	-25/30	60	70.7	94.1			

3. Test program for strand-type GFRP specimens

3.1. Specimen preparation and testing method

In order to examine the degradation of GFRP composite in micro level, strand-type E-glass/vinylester rod specimens were fabricated and tested. As shown in Fig. 7, the size of strand-type specimen is much smaller than that of the full-size specimen. The specimen of 0.7 mm in diameter was pultruded in the laboratory and cured for 24 h at room temperature, 2 h at 70 °C, and 2 h at 120 °C. The fiber volume fraction of the specimen was approximately 45% by weight.

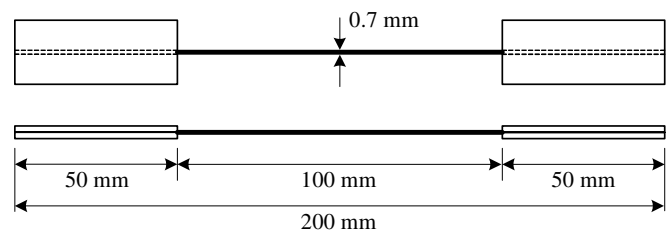


Fig. 7. Dimensions of strand-type specimen.

The E-glass fiber used in this study was ER270FW manufactured by Hankuk Fiber Co. in Korea, it has a diameter of 13 μm , a density of 2.54 g/cm^3 , a tensile strength of

Table 7
Aging methods for strand-type specimens

Conditions	Solution	Temperature (°C)	Duration (days)
Moisture	Distilled water	20, 80	1, 2, 5, 10, 20, 30, 60, 120, 150
Alkali	Alkaline solution	20, 80	1, 2, 5, 10, 20, 30, 60, 120, 150
Chloride	3% NaCl solution	20	2, 10, 30, 70
Deicing agent	4% CaCl ₂ solution	20	2, 10, 30, 70
Wetting–drying	Distilled water	20	2, 10, 30, 70

2.7 GPa, and an elastic modulus of 77 GPa. The vinyl ester resin was RF-1001 manufactured by Cray Valley Korea Inc., and 1.0% of methyl ethyl ketone peroxide and 0.4% of cobalt chloride by weight were used as a catalyst and accelerator, respectively.

The aging methods used for the strand-type specimens are summarized in Table 7. The solutions used to simulate the alkaline and chloride conditions are the same as those in Section 2.1. Four-percent calcium chloride (CaCl₂) solution was used to simulate the condition due to a deicing agent. A wetting and drying cycle consisted of two days of immersion at 20 °C followed by two days of drying at 60 °C.

A total of 520 specimens were exposed to five different environmental conditions. The specimens prepared for each environmental condition were placed in a beaker that was filled with each solution. At the end of the designated immersion times, a weight gain of the conditioned specimen was measured and tensile test was performed to measure the tensile strength.

Fig. 8 shows a test set-up for tensile test using a UTM of 5 kN capacity with air pressure chucks. A pair of 20 × 50 mm cardboard paper tabs was adhesively bonded

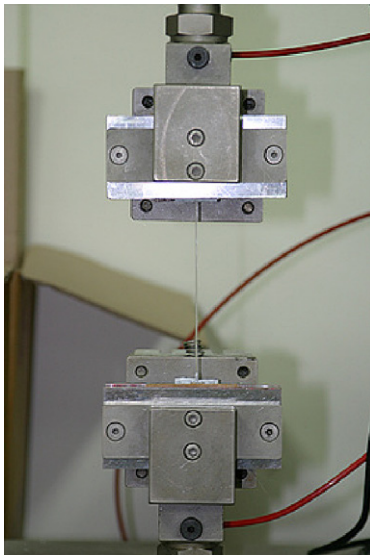


Fig. 8. Test set-up for tensile test.

to each end of the specimen for the test. The gauge length of the strand-type specimen was about 100 mm. The specimen was loaded under displacement control with loading rate of 2.0 mm/min until the failure occurred. The average tensile strength of unconditioned strand-type specimen was 2.77 GPa.

3.2. Test results and discussion

The fluid absorption property of an FRP composite can be evaluated by weight gain measurement. Fig. 9 shows the weight gain versus the square root of immersion time curves for the specimens immersed in the water and alkaline solution. The diffusivity of the specimens was roughly linear for the given immersion time and was greatly affected by temperature. The weight gain of the specimens conditioned at 80 °C was approximately three times greater than those conditioned at 20 °C. For both the temperatures, the weight gain measured for the specimens immersed in alkaline solution was slightly lower than those in water.

Fig. 10 shows the tensile strength versus the immersion time curves for the specimens immersed in water and

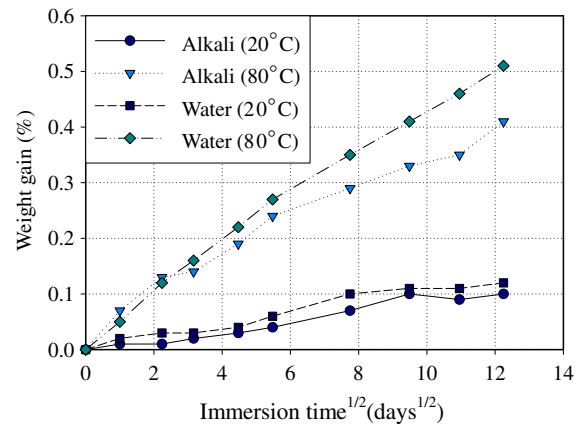


Fig. 9. Weight gain of strand-type specimens immersed in alkaline solution and water.

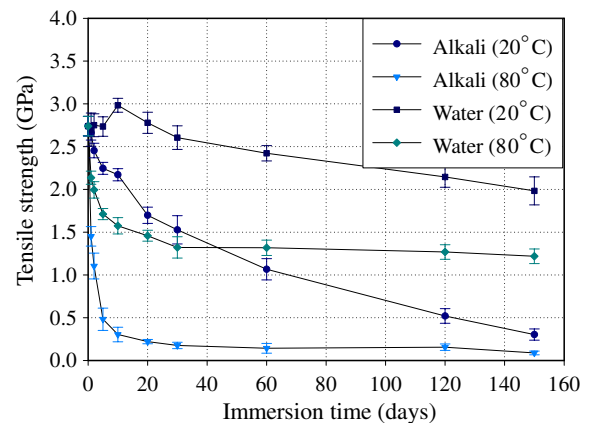


Fig. 10. Variation of tensile strength of strand-type specimens immersed in water and alkaline solution.

alkaline solution. Although the diffusivity was similar for both conditions, the tensile strength of the specimens immersed in alkaline solution significantly decreased as the immersion time and temperature increased.

The average strength retention of the specimens immersed in alkaline solution at 20 °C and 80 °C for 150 days was 11.1% and 3.2%, respectively. On the other hand, the average strength retention of the specimens immersed in water at 20 °C and 80 °C for 150 days was 72.4% and 44.5%, respectively. The test results indicated that the alkaline solution severely caused the degradation of E-glass/vinylester composite.

Fig. 11 shows the tensile strength versus the immersion time curves for the specimens exposed to other environ-

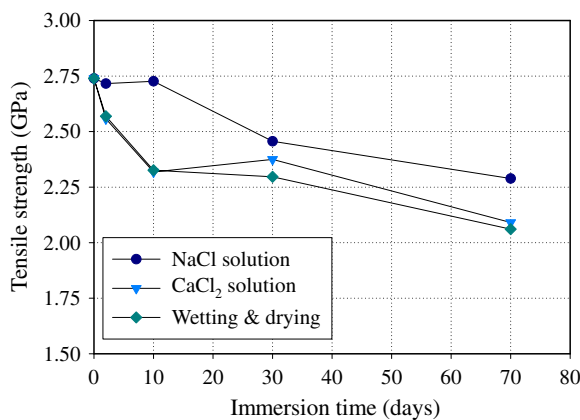


Fig. 11. Variation of tensile strength of strand-type specimens exposed to other environmental conditions.

mental conditions. In the figure, the strength retention of the specimens immersed in NaCl and CaCl₂ solutions for 70 days was 83.6% and 76.3%, respectively. The strength retention of the specimens exposed to wetting and drying cycles for 70 days was 75.2%. The influence of other environmental conditions on the tensile strength of the specimens was not significant and their influences were similar to that of the water at 20 °C.

To examine the physical conditions of the specimens exposed to the environmental conditions in micro level, typical SEM images were taken at fractured section after the tensile test. Fig. 12 shows a series of SEM images for the fractured section of the specimens. For the unconditioned specimen shown in Fig. 12(a), many ‘hackles’ can be observed at the fiber surfaces that are the matrix bonded to the fibers even after the fracture. This verifies that there is a strong bond at the interface between the fiber and the matrix before the degradation. On the other hand, as the temperature and immersion time increased, less ‘hackles’ can be seen at the fiber surfaces for the specimens immersed in the water, as shown in Fig. 12(b)–(d).

Fig. 13 shows a series of SEM images for the fractured section of the specimens immersed in alkaline solution. For the specimen conditioned at 20 °C for 5 days, as shown in Fig. 13(a), the relatively clean fiber surfaces resulted from the weak fiber–matrix bonding can be observed. As the temperature and immersion time further increased, the matrix bonded to the fibers could hardly be identified (Fig. 13(b)–(d)).

Through the aid of SEM images, it turned out that water and alkali in GFRP degrade not only glass fiber

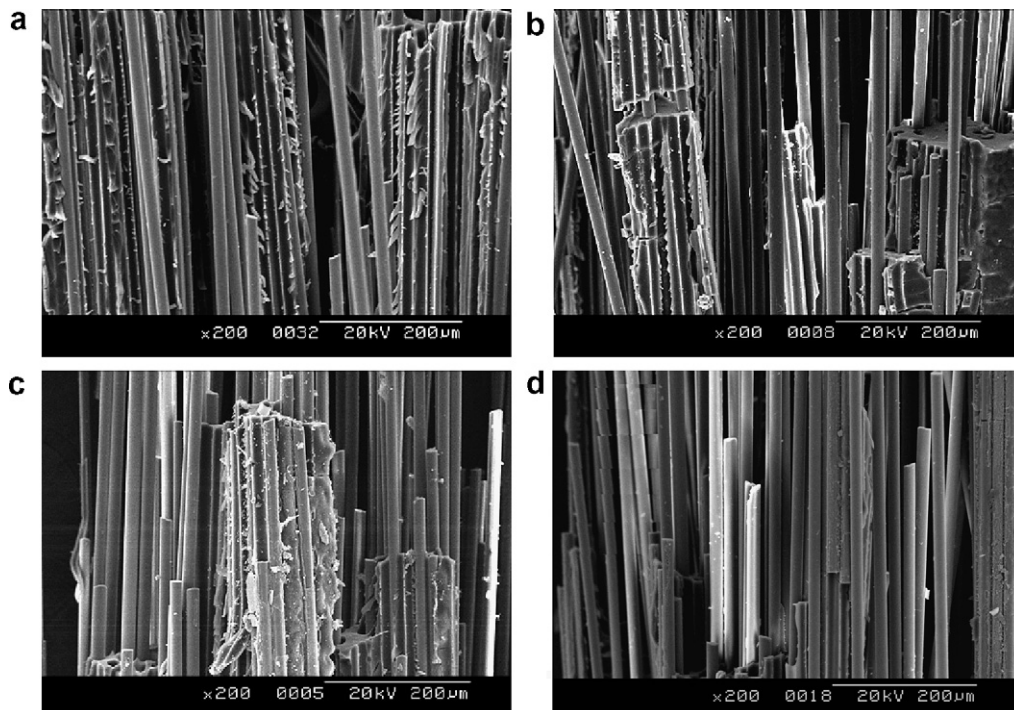


Fig. 12. SEM images for specimens: (a) unconditioned; (b) water at 20 °C for 60 days; (c) water at 80 °C for 60 days; and (d) water at 80 °C for 150 days.

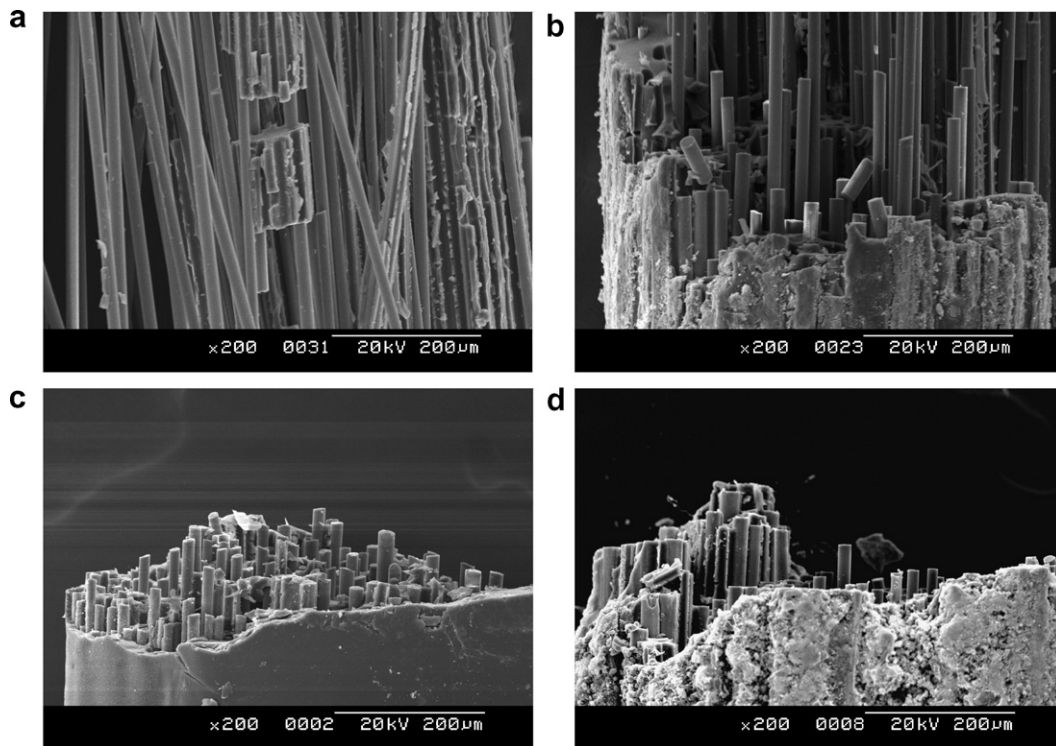


Fig. 13. SEM images for specimens immersed in alkaline solution: (a) 20 °C for 5 days; (b) 80 °C for 5 days; (c) 20 °C for 150 days; and (d) 80 °C for 150 days.

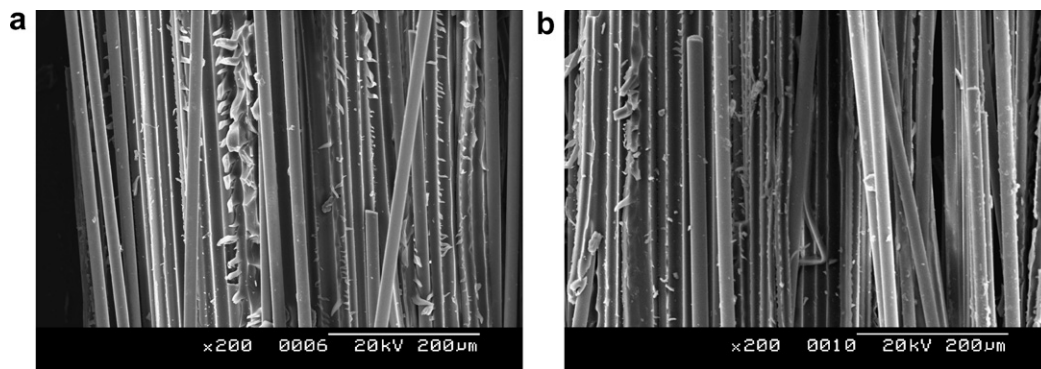


Fig. 14. SEM images for specimens immersed in (a) NaCl solution and (b) CaCl_2 solution.

and matrix but also the interface between the fiber and the matrix. It is believed that the reduction in bond strength has a strong correlation with the reduction in tensile strength of GFRP.

Fig. 14 shows a series of SEM images for the fractured section of the specimens exposed to the other environmental conditions for 70 days. The SEM images of the fractured section indicated good bonding between the fiber and the matrix at the interface.

4. Summary and conclusions

In this study, E-glass/vinylester rods were tested by using the two levels of short-term durability testing proto-

col. The findings of this study along with recommendation for future study are discussed in the following.

The results of the short-term durability test for the E-glass/vinylester rods indicated that the tensile properties of the rods are significantly reduced under all the environmental conditions considered in this study. Overall, alkaline environmental condition had more influence on the degradation of GFRP rods than the other influencing factors. The results of this test confirmed the results of the existing studies that moisture and alkali significantly affect not only the degradation of glass fiber and matrix but the interface between the fiber and the matrix.

To ensure long-term durability of FRP rebars in civil infrastructure applications, the individual and combined

effects of environmental conditions on FRP rebar itself as well as on the interface between rebar and concrete should be accessed. Furthermore, to guarantee the performance of FRP rebars during the typical civil infrastructure service life, actual behavior of concrete members reinforced with FRP rebars under the actual service loading and environmental conditions must be investigated. One of major tasks in the future study for the durability of FRP bars would be an assessment of the durability performance of FRP rebar reinforced structures and elements under service loading and natural weathering conditions.

The existing studies have utilized a full-size test specimen. Although many different types of gripping mechanisms have been introduced for tensile testing of FRP rods, gripping of FRP rod specimens is difficult task. Furthermore, the conditioning of full-size specimen generally requires larger size of equipments.

The usability of a strand-type specimen in durability test for FRP composites was investigated in this study. The conditioning and testing of the strand-type specimen was much easier than the conventional ones. The use of this testing method may be ideal for the repetitive test such as long-term durability test. However, one of the limitations using the strand-type specimen is that the interfacial shear strength cannot be tested. In order to utilize this testing method to predict long-term durability of prototype FRP rods, an effective method to correlate the test results using the strand-type specimens with full-size specimens should be developed.

Acknowledgements

This study was funded by the Korea Research Council of Public Science and Technology. The authors would like to thank Drs. Julio F. Davalos and Indrajit Ray, West Virginia University in the USA, for the information on test protocol for short-term durability test of composite materials.

References

- [1] Karbhari VM, Zhao L. Use of composites for 21st century civil infrastructure. *Comput Methods Appl Mech Engrg* 2000;185: 433–54.
- [2] Bakis CE, Bank LC, Brown VL, Cosenza E, Davalos JF, Lesko JJ, et al. Fiber-reinforced polymer composites for construction-state-of-the-art review. *ASCE J Compos Constr* 2002;6(2):73–87.
- [3] El-Salakawy E, Benmokrane B, El-Ragaby A, Nadeau D. Field investigation on the first bridge deck slab reinforced with glass FRP bars constructed in Canada. *ASCE J Compos Constr* 2005;9(6): 470–9.
- [4] Benmokrane B, El-Salakawy E, El-Ragaby A, Lackey T. Designing and testing of concrete bridge decks reinforced with glass FRP bars. *ASCE J Bridge Engrg* 2006;11(2):217–29.
- [5] Federal Highway Administration (FHWA). Standard database reports: IBRC database; 2006. <http://www.fhwa.dot.gov/bridge/ibrc/resource.cfm> [Updated 22.9.06].
- [6] Gangarao HVS, Vijay PV. Aging of structural composites under varying environmental conditions. In: *Proceedings of the third international symposium on non-metallic (FRP) reinforcement for concrete structures (FRPRCS-3)*, Sapporo, Japan, October 14–16, vol. 2. 1997, p. 91–8.
- [7] Uomoto T. Durability design of GFRP rods for concrete reinforcement. In: *Proceedings of the sixth international symposium on FRP reinforcement for concrete structures (FRPRCS-6)*, Singapore, July 8–10, 2003. p. 37–50.
- [8] Deje V. Durability of FRP reinforcement in concrete. Thesis for the degree of Licentiate of Engineering. Sweden: Chalmers University of Technology; 2001.
- [9] Karbhari VM, Chin JW, Hunston D, Benmokrane B, Juska T, Morgan R, et al. Durability gap analysis for fiber-reinforced polymer composites in civil infrastructure. *ASCE J Compos Constr* 2003;7(3):238–47.
- [10] Liao K, Schultheisz CR, Hunston DL. Effects of environmental aging on the properties of pultruded GFRP. *Composites B* 1999;30: 485–93.
- [11] Chu W, Wu L, Karbhari VM. Durability evaluation of moderate temperature cured E-glass/vinylester systems. *Comp Struct* 2004;66:367–76.
- [12] Nkurunziza G, Debaiky A, Cousin P, Benmokrane B. Durability of GFRP bars: a critical review of the literature. *J Prog Struct Engrg Mater* 2005;7:194–209.
- [13] Micelli F, Nanni A. Durability of FRP rods for concrete structures. *Constr Build Mater* 2004;18:491–503.
- [14] Chen Y, Davalos JF, Ray I, Kim H-Y. Accelerated aging tests for evaluations of durability performance of FRP reinforcing bars for concrete structures. *Comp Struct* 2007;78:101–11.
- [15] Abbasi A, Hogg PJ. Temperature and environmental effects on glass fibre rebar: modulus, strength and interfacial bond strength with concrete. *Composites B* 2005;36:394–404.
- [16] Abdel-Magid B, Ziaee S, Gass K, Schneider M. The combined effects of load, moisture and temperature on the properties of E-glass/epoxy composites. *Comp Struct* 2005;71:320–6.
- [17] US Army Corps of Engineers. Investigations of plastic composite materials for highway safety structures. Cold Regions Research and Engineering Laboratory (CRREL). Report 98-7; 1998. p. 41–6.
- [18] Benmokrane B, Wang P, Ton-That TM, Rahman H, Robert J-F. Durability of glass fiber-reinforced polymer reinforcing bars in concrete environment. *ASCE J Compos Constr* 2002;6(3): 143–53.
- [19] Nkurunziza G, Benmokrane B, Debaiky AS, Masmoudi R. Effect of creep and environment on long-term tensile properties of glass FRP reinforcing bars. In: *Proceedings of the fourth international conference on advanced composite materials in bridges and structures (ACMBS)*, Calgary, Canada, July 20–23, 2004. p. 1–8.
- [20] Al-Zahrani MM, Al-Dulaijan SU, Nanni A, Bakis CE, Boothby TE. Evaluation of bond using FRP rods with axisymmetric deformations. *Constr Build Mater* 1999;13:299–309.
- [21] Lorenzis LD, Rizzo A, Tegola AL. A modified pull-out test for bond of near-surface mounted FRP rods in concrete. *Composites B* 2002;33:589–603.
- [22] Katz A, Berman N, Bank LC. Effect of high temperature on bond strength of FRP rebars. *ASCE J Compos Constr* 1999;3(2):73–81.
- [23] Katz A. Bond to concrete of FRP rebars after cyclic loading. *ASCE J Compos Constr* 2000;4(3):137–44.
- [24] Alsayed SH. Flexural behaviour of concrete beams reinforced with GFRP bars. *Cem Concr Compos* 1998;20:1–11.
- [25] Alsayed SH, Al-Salloum YA, Almusallam TH. Performance of glass fiber reinforced plastic bars as a reinforcing material for concrete structures. *Composites B* 2000;31:555–67.
- [26] Abdalla HA. Evaluation of deflection in concrete members reinforced with fibre reinforced polymer (FRP) bars. *Comp Struct* 2002;56:63–71.
- [27] Toutanji H, Deng Y. Deflection and crack-width prediction of concrete beams reinforced with glass FRP rods. *Constr Build Mater* 2003;17:69–74.
- [28] Laoubi K, El-Salakawy E, Benmokrane B. Creep and durability of sand-coated glass FRP bars in concrete elements under freeze/

- thaw cycling and sustained loads. *Cem Concr Compos* 2006;28: 869–78.
- [29] Almusallam TH, Al-Salloum YA. Durability of GFRP rebars in concrete beams under sustained loads at severe environments. *J Compos Mater* 2006;40(7):623–37.
- [30] ASTM D 3916. Standard test method for tensile properties of pultruded glass–fiber–reinforced plastic rods; 2002.
- [31] ASTM D 4475. Standard test method for apparent shear strength of pultruded reinforced plastic rods by the short-beam method; 2002.